

Statistical Models in Computational Biology

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Please submit your project with the filename Lastname(s)_Project11.pdf.

Problem 27: AR(1) model**(2 points)**

Consider a first-order autoregressive process AR(1) specified by the conditional distribution

$$P(X^t = x^t \mid X^{t-1} = x^{t-1}) = \text{Norm}(x^t \mid Ax^{t-1}, \Gamma),$$

where $X^t \in \mathbb{R}^n$, $A \in \mathbb{R}^{n \times n}$, and $\Gamma \in \mathbb{R}^{n \times n}$ is an identity matrix. Given a realization $x^1, \dots, x^T$, show that the log-likelihood is

$$\ell(A) = -\frac{1}{2} \sum_{t=1}^T (x^t - Ax^{t-1})^\top (x^t - Ax^{t-1}).$$

Derive the score equation for A and compute $\hat{A}_{\text{MLE}}$.Hint: You can use trace identity $u^\top u = \text{tr}(u^\top u) = \text{tr}(uu^\top)$, where $u \in \mathbb{R}^n$.**Problem 28: Modeling overdispersion in RNA-seq gene expression****(8 points)**

In an RNA-seq experiment, we measure gene expression by counting the number of reads mapped to each gene across n samples (e.g. cells). This is an important step in count-based differential expression analysis, used for identifying genes whose expression levels differ significantly between biological conditions.

We will restrict our analysis to a single gene throughout this exercise. For a given gene, let Y_i denote the read count observed in sample i . A naive model assumes the Poisson distribution, $Y_i \sim \text{Pois}(\mu_i)$, where μ_i is the expected read count in sample i . However, in practice, read counts exhibit overdispersion – the observed variance across samples is much larger than the mean – due to two sources of variability:

- Technical noise: amplification, reverse transcription, and library preparation. In this exercise, we will make a simplifying assumption and model this type of noise using the Poisson model.
- Biological variability: sample-to-sample differences in expression levels, which inflate the variance beyond the Poisson prediction.

To account for both sources of variation, we adopt a hierarchical model. Let Z_i denote the latent (unobserved) expression rate of sample i , which varies randomly across samples due to biological variability. Conditionally on $Z_i = z$, the observed count follows the Poisson distribution:

$$Y_i \mid Z_i = z \sim \text{Pois}(z),$$

and the latent variable itself follows the Gamma distribution representing biological heterogeneity:

$$Z_i \sim \text{Gamma}(\alpha_i, \nu_i), \quad E(Z_i) = \mu_i = \alpha_i \nu_i, \quad \text{Var}(Z_i) = \alpha_i^2 \nu_i$$

where α_i and ν_i are scale and shape parameters of the Gamma distribution, respectively.

1. (1 point) Show that the marginal mean and variance of the read count Y_i are

$$E(Y_i) = \mu_i = \alpha_i \nu_i, \quad \text{Var}(Y_i) = \alpha_i \nu_i + \alpha_i^2 \nu_i,$$

confirming that $\text{Var}(Y_i) > E(Y_i)$, i.e. the counts are overdispersed. Interpret the two terms in $\text{Var}(Y_i)$: which term, according to our model assumptions, corresponds to technical noise and which to biological variability?

Hint 1. You can use the Law of Total Expectation and Law of Total Variance, namely, $E[Y] = E[E[Y | Z]]$ and $\text{Var}(Y) = E[\text{Var}(Y | Z)] + \text{Var}(E[Y | Z])$.

2. (1 point) Suppose that Y_i are independent across samples with μ_i not all equal (e.g. cells come from two experimental conditions – before and after the treatment). Model **A** for differential analysis of count data assumes a shared known shape parameter of the Gamma distribution, $\nu_i = \nu$, across all samples. Show that under this assumption the distribution of Y_i has the form of the exponential dispersion family with $\phi = 1$, and show that the variance function is

$$V(\mu) = \mu + \frac{\mu^2}{\nu},$$

which is quadratic in μ .

Hint 1: You can use the fact that marginalising over Z_i gives the negative binomial distribution for Y_i :

$$P(Y_i = y) = \binom{y + \nu_i - 1}{y} \left(\frac{\mu_i}{\nu_i + \mu_i} \right)^y \left(\frac{\nu_i}{\nu_i + \mu_i} \right)^{\nu_i}$$

Hint 2. Recall that the exponential dispersion family (EDF) has the form:

$$\log p(y; \theta, \phi) = \frac{y\theta - b(\theta)}{\phi} + c(y, \phi),$$

with mean $\mu = b'(\theta)$ and variance $\text{Var}(Y) = \phi V(\mu)$, where $V(\mu) = b''(\theta)$ is the variance function and $\phi > 0$ is the dispersion parameter.

3. (3 points) Model **B** instead assumes a shared known scale parameter of the Gamma distribution, $\alpha_i = \alpha$, across samples. Show that in this case the variance function is

$$V(\mu) = \mu,$$

with dispersion parameter $\phi = 1 + \alpha$, so that the total variance is:

$$\text{Var}(Y_i) = \phi V(\mu) = (1 + \alpha)\mu,$$

which is linear in μ . Show further that the distribution of Y_i does not belong to the exponential dispersion family, more specifically, the log-likelihood cannot be written as $\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)$ with $c(y, \phi)$ independent of μ .

Write down the quasi-likelihood for observation y_i .

Hint 1: Recall that the quasi-likelihood can be calculated using $Q_i(\mu_i; y_i) = \int_{y_i}^{\mu_i} \frac{y_i - t}{\phi \cdot V(t)} dt$. You should write down the expression after performing the integration.

4. (3 points) In Models **A** and **B**, the dispersion structure is rigid: one parameter is fixed and the other is completely determined by μ_i through the constraint $\alpha_i \nu_i = \mu_i$, giving either a quadratic variance function $V(\mu) = \mu + \mu^2/\nu$ (Model **A**) or a linear variance function $V(\mu) = \mu$ with constant overdispersion $\phi = 1 + \alpha$ (Model **B**).

Now suppose a more general case, where both parameters are allowed to vary with :

$$\alpha_i = \gamma + \psi \mu_i, \quad \nu_i^{-1} = \psi + \gamma \mu_i^{-1},$$

which is consistent with $\alpha_i \nu_i = \mu_i$. Show that the variance function becomes

$$V(\mu) = \mu + \gamma \mu + \psi \mu^2,$$

which takes an over-dispersed Poisson form when $\psi = 0$ or the negative binomial form when $\gamma = 0$. Show that Y_i does not have the exponential dispersion family form, and write down the quasi-likelihood (see *Hint 1* in Part 3).

Hint 2: Notice that the integral can be expressed as a difference of two integrals namely,

$$\int_{y_i}^{\mu_i} \frac{y_i}{t(1+\gamma+\psi t)} dt = \int_{y_i}^{\mu_i} \frac{y_i}{t(1+\gamma)} dt - \int_{y_i}^{\mu_i} \frac{y_i \psi}{(1+\gamma)(1+\gamma+\psi t)} dt$$